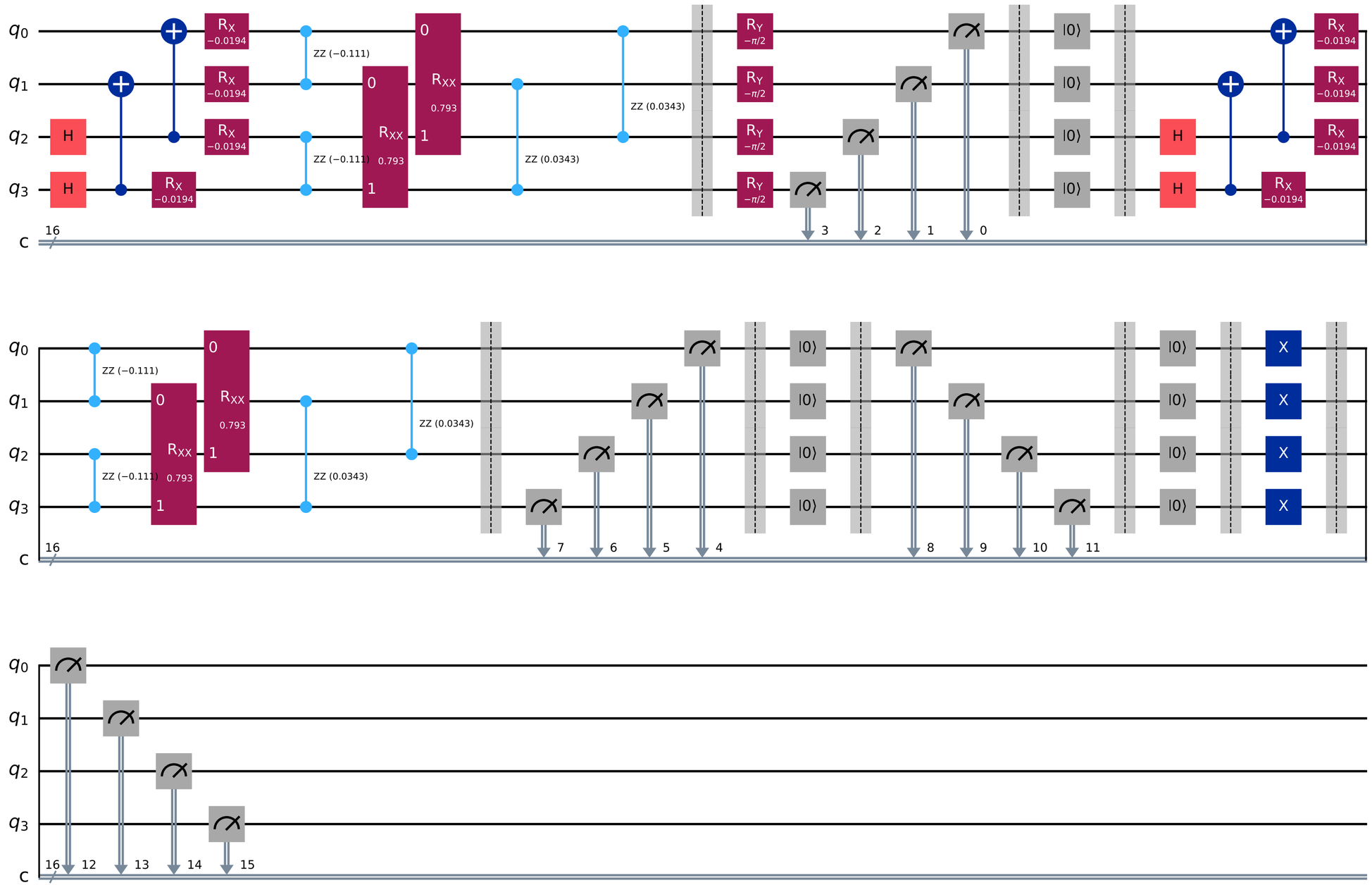
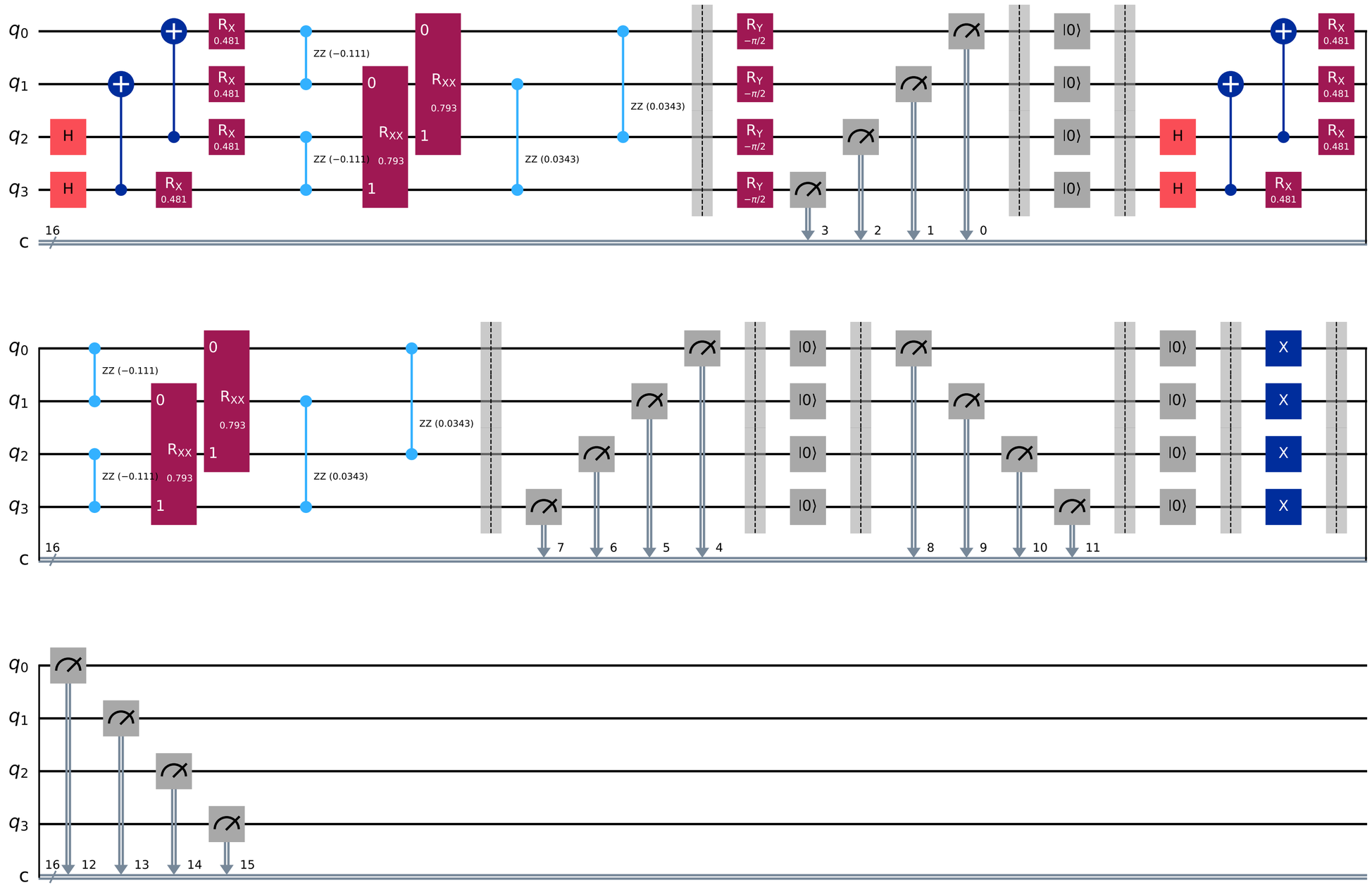
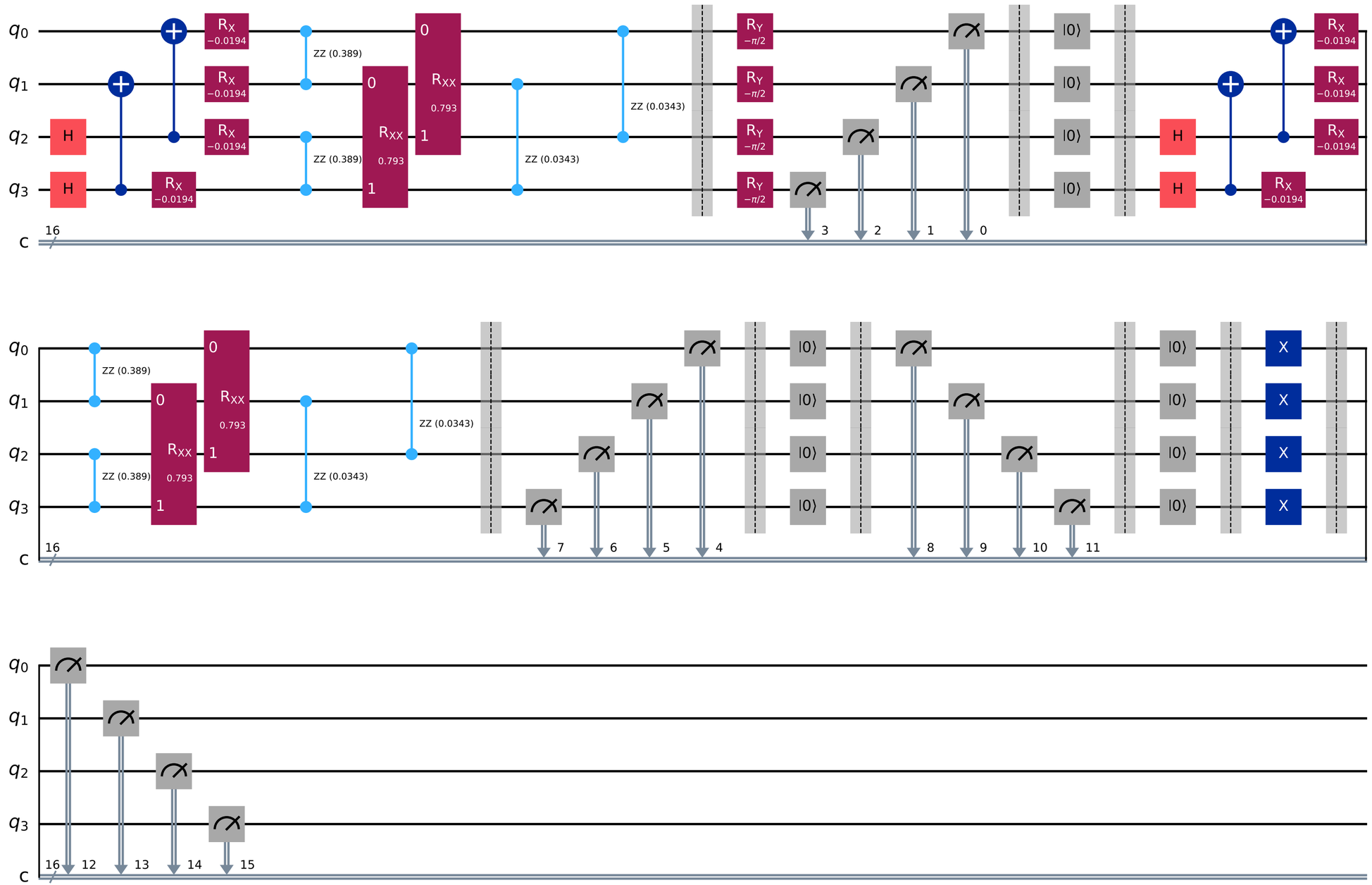
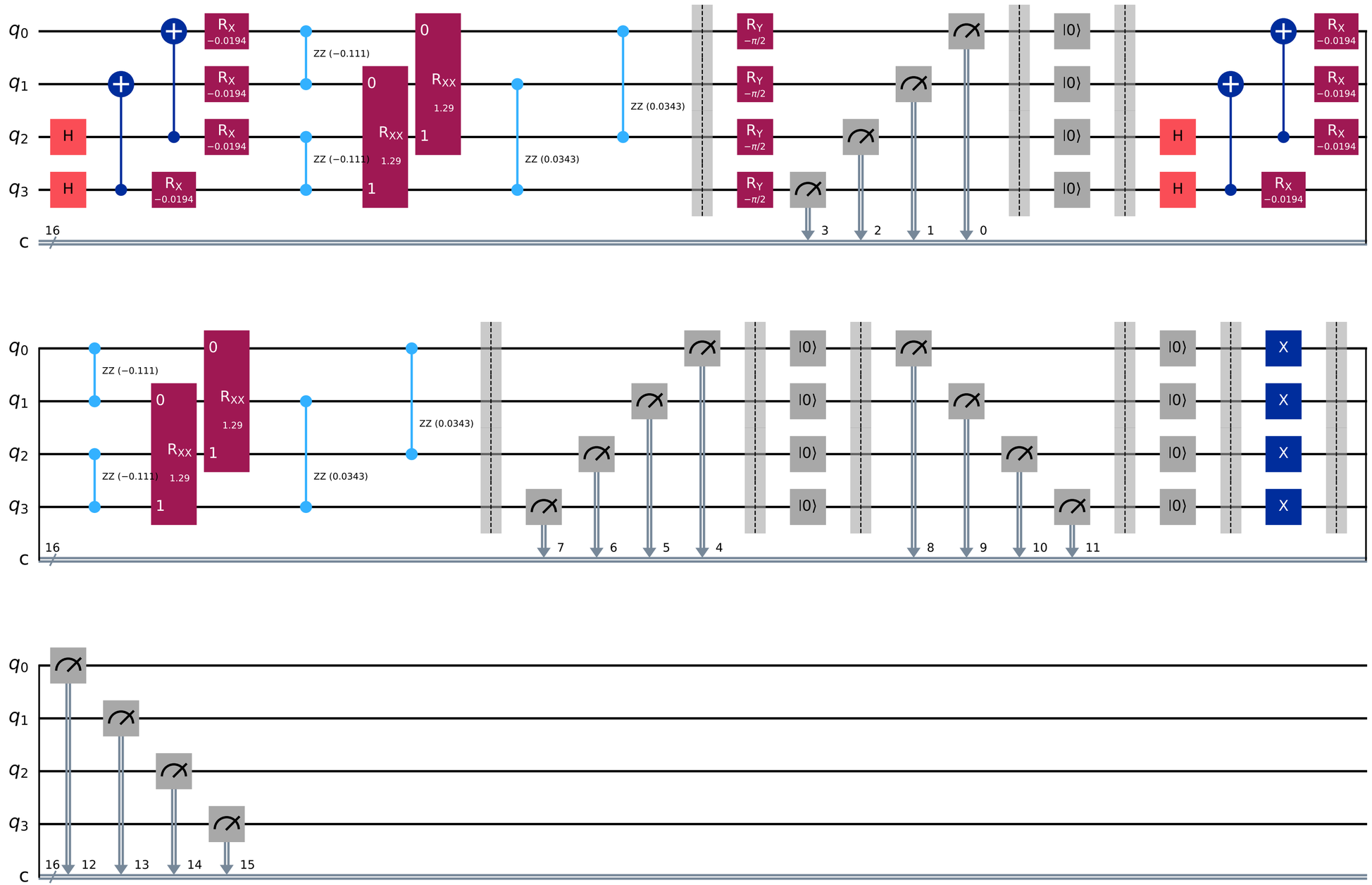
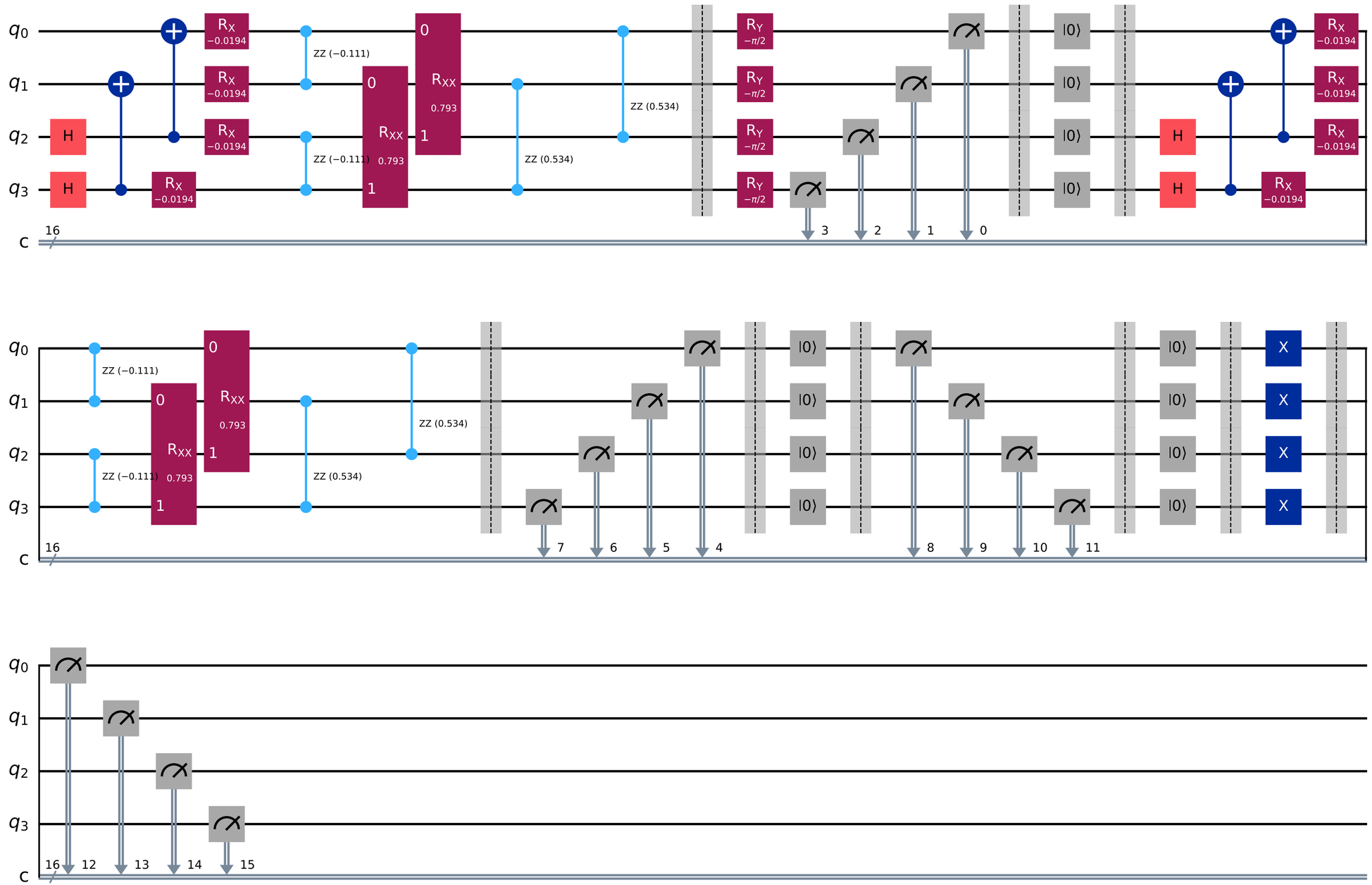
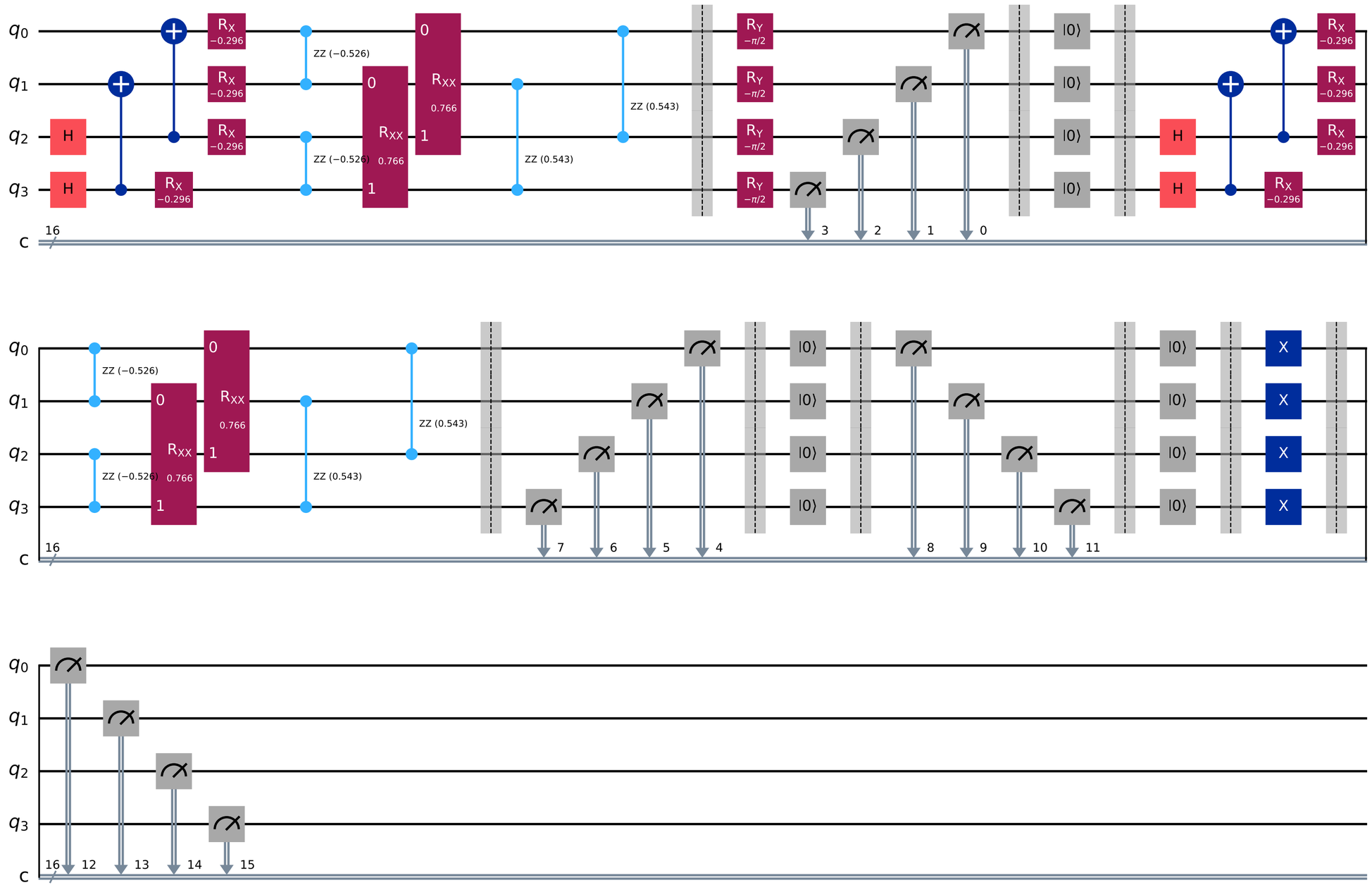

q0
q1
q2
q3
c
16
H
H
RX −0.0194
RX −0.0194
RX −0.0194
RX −0.0194
ZZ (−0.111)
ZZ (−0.111)
0
RXX
0.793
1
0
RXX
0.793
1
ZZ (0.0343)
ZZ (0.0343)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0194
RX −0.0194
RX −0.0194
RX −0.0194
ZZ (−0.111)
ZZ (−0.111)
0
RXX
0.793
1
0
RXX
0.793
1
ZZ (0.0343)
ZZ (0.0343)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

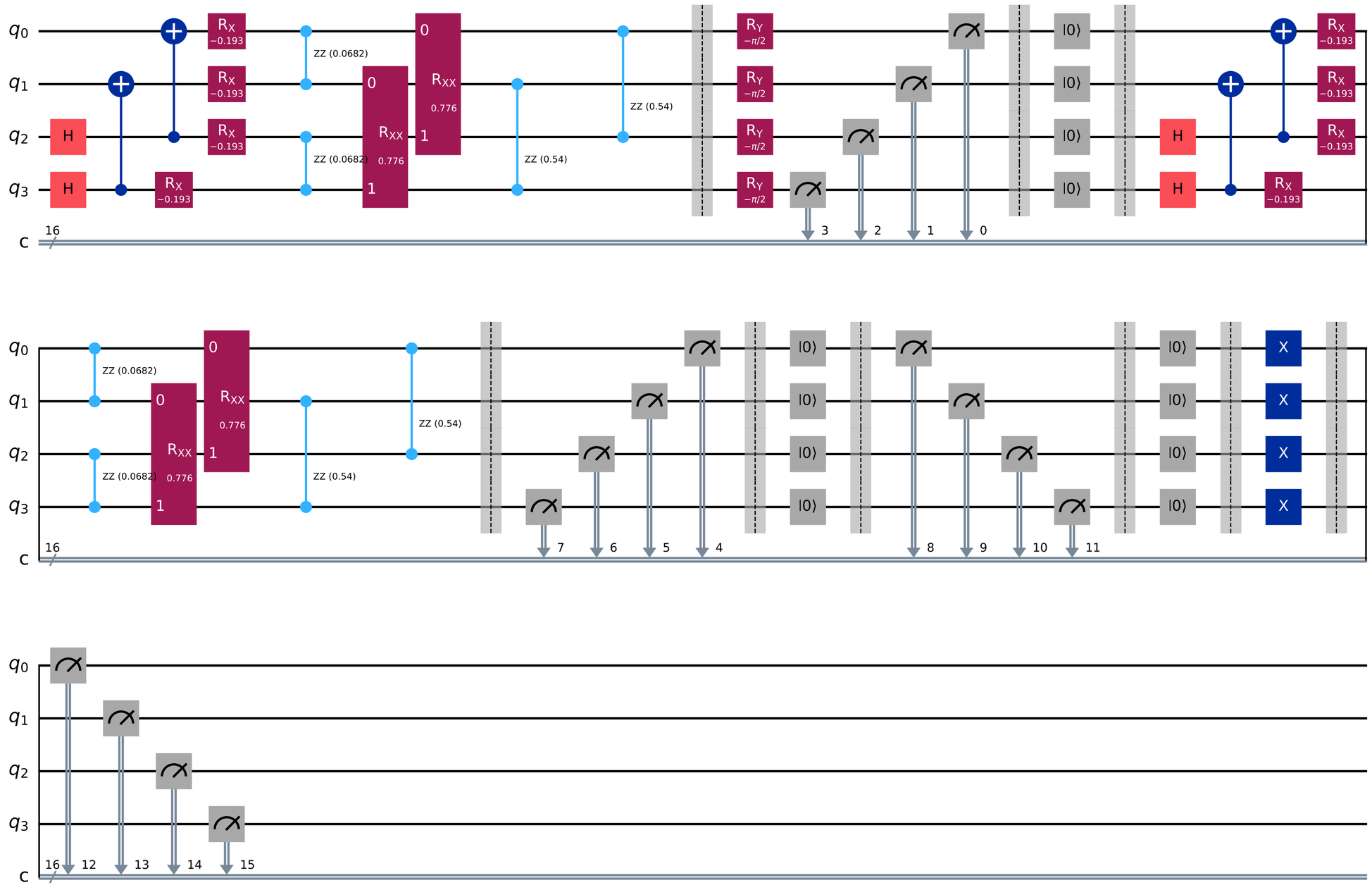

q0
q1
q2
q3
c
16
H
H
RX 0.481
RX 0.481
RX 0.481
RX 0.481
ZZ (−0.111)
ZZ (−0.111)
RXX 0.793
ZZ (0.0343)
ZZ (0.0343)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.481
RX 0.481
RX 0.481
RX 0.481
ZZ (−0.111)
ZZ (−0.111)
RXX 0.793
ZZ (0.0343)
ZZ (0.0343)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

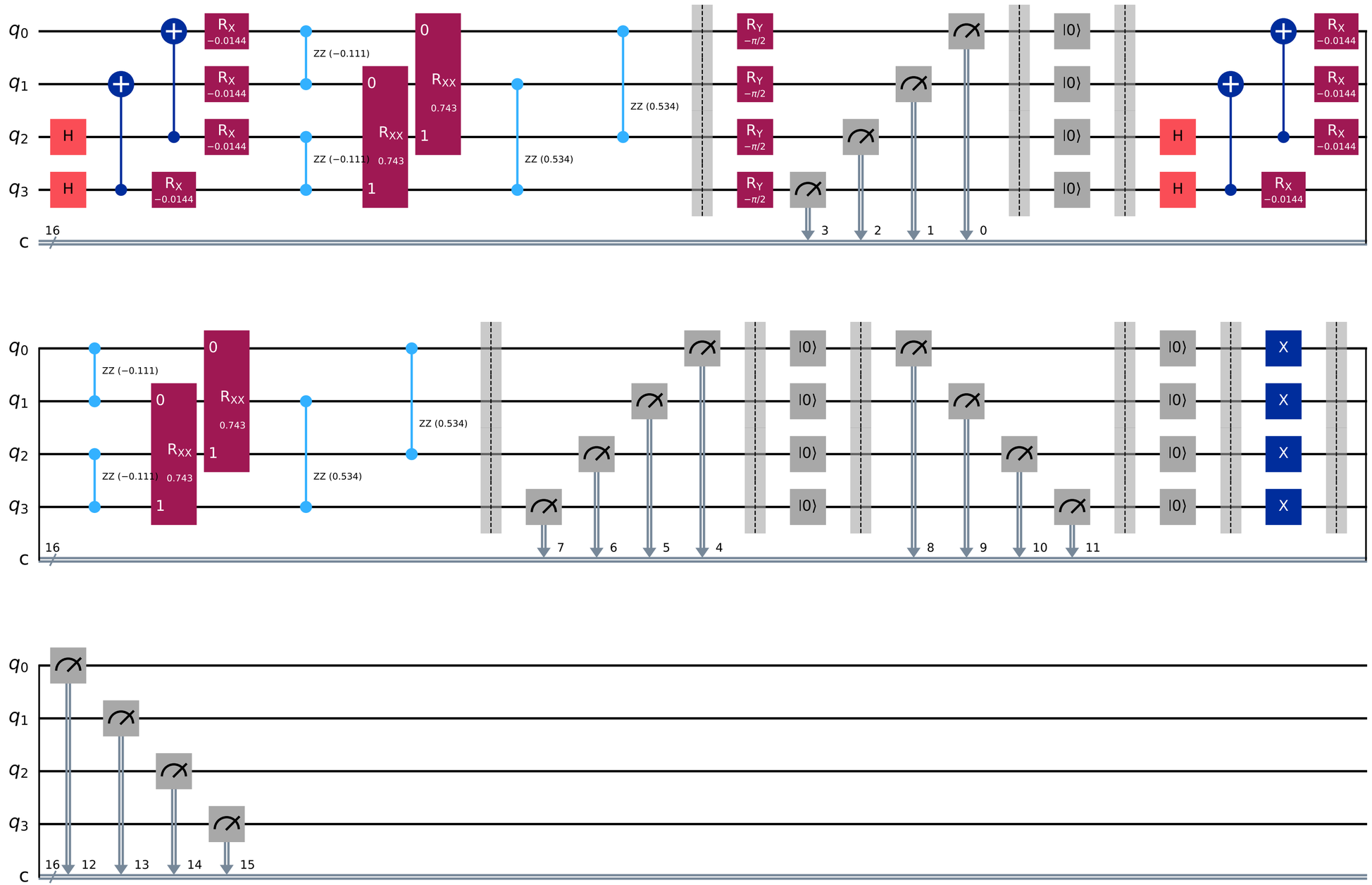

q0
q1
q2
q3
c
16
H
H
RX −0.0194
RX −0.0194
RX −0.0194
RX −0.0194
ZZ (0.389)
ZZ (0.389)
0
RXX 0.793
1
0
RXX 0.793
1
ZZ (0.0343)
ZZ (0.0343)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0194
RX −0.0194
RX −0.0194
RX −0.0194
ZZ (0.389)
ZZ (0.389)
0
RXX 0.793
1
0
RXX 0.793
1
ZZ (0.0343)
ZZ (0.0343)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

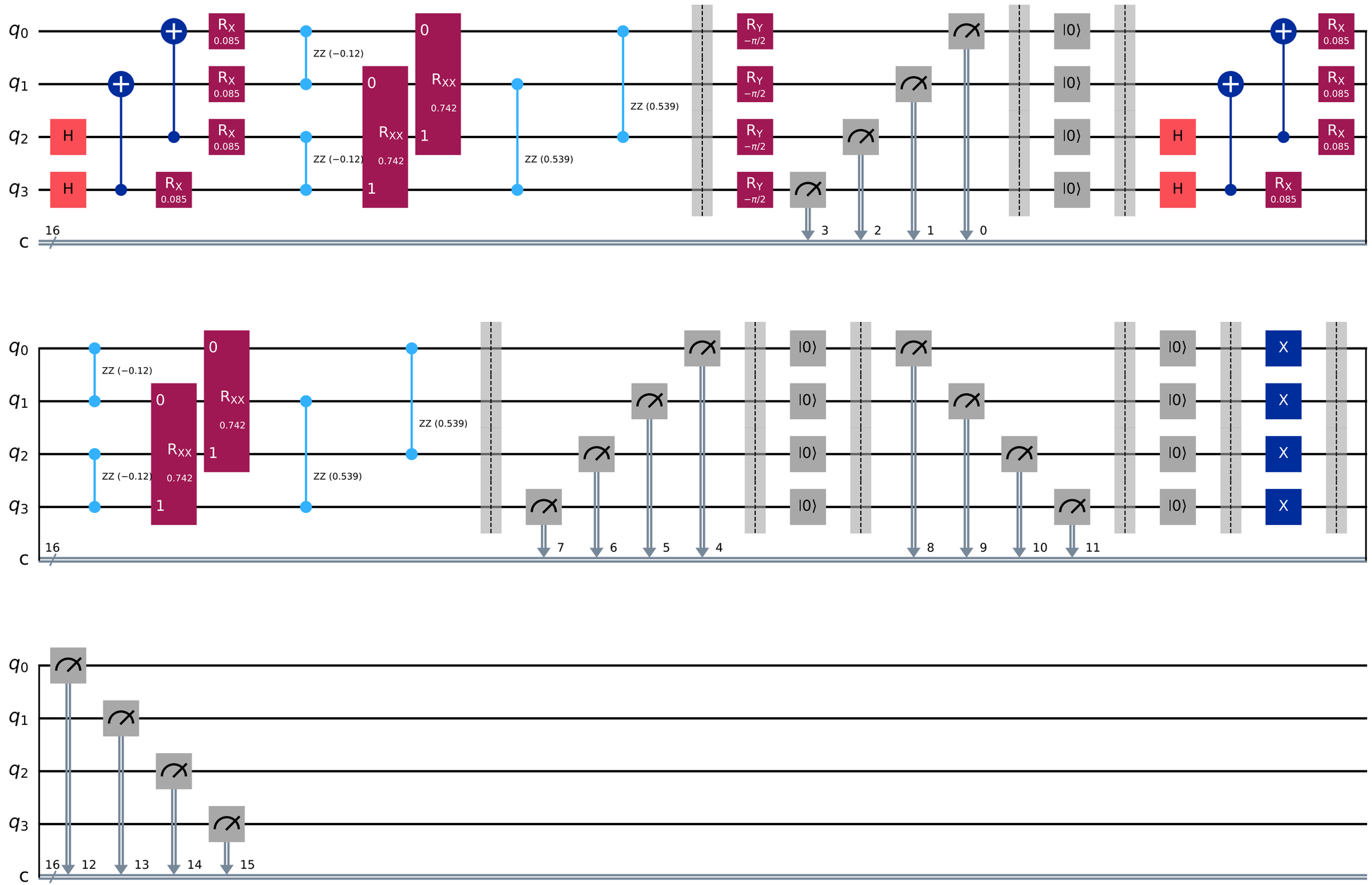

q0
q1
q2
q3
c
16
H
H
RX −0.0194
RX −0.0194
RX −0.0194
RX −0.0194
ZZ (−0.111)
ZZ (−0.111)
0
RXX 1.29
1
ZZ (0.0343)
ZZ (0.0343)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0194
RX −0.0194
RX −0.0194
RX −0.0194
ZZ (−0.111)
ZZ (−0.111)
0
RXX 1.29
1
ZZ (0.0343)
ZZ (0.0343)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

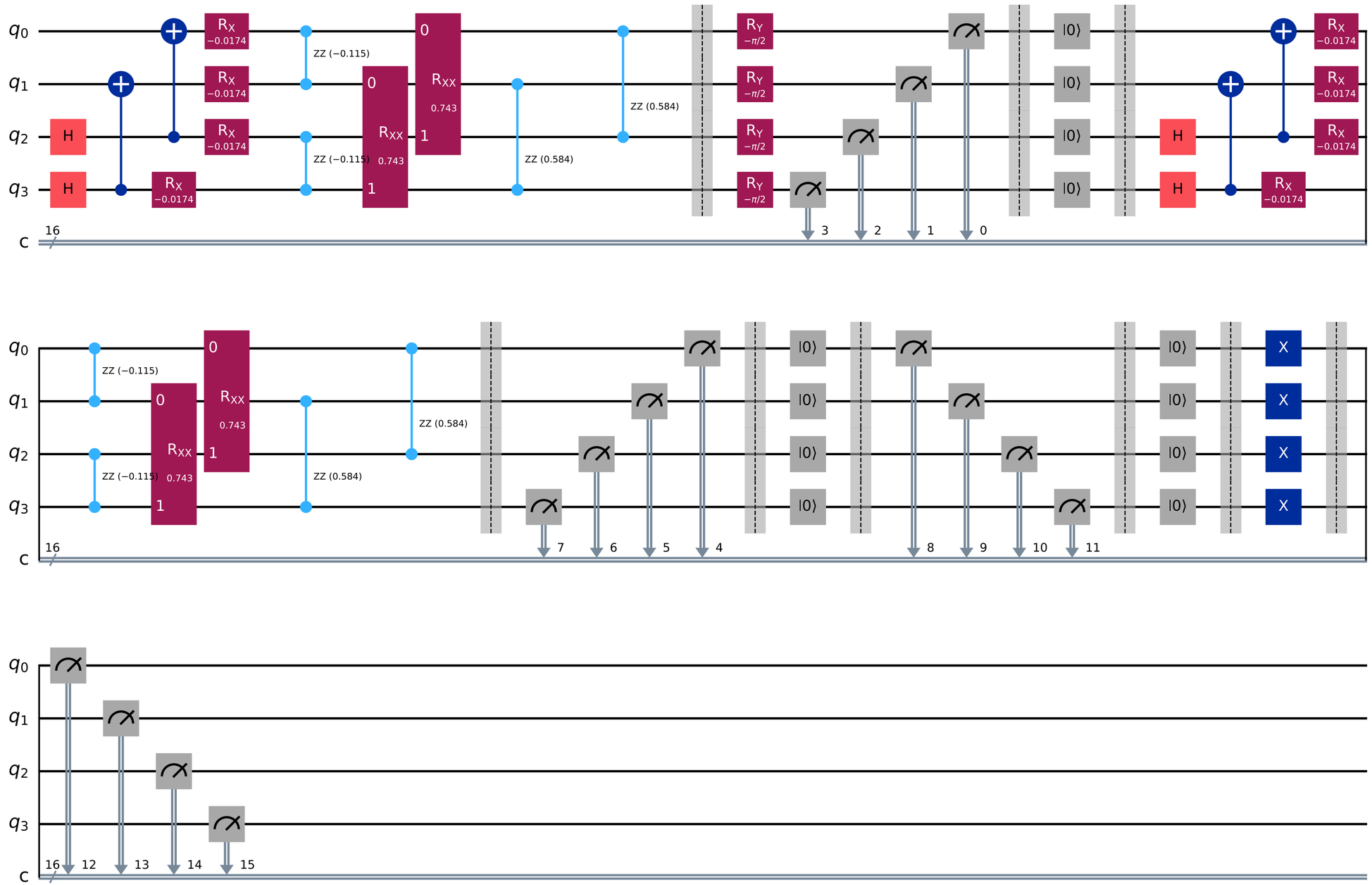

q0
q1
q2
q3
c
16
H
H
RX −0.0194
RX −0.0194
RX −0.0194
RX −0.0194
ZZ (−0.111)
ZZ (−0.111)
0
RXX 0.793
1
0
RXX 0.793
1
ZZ (0.534)
ZZ (0.534)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0194
RX −0.0194
RX −0.0194
RX −0.0194
ZZ (−0.111)
ZZ (−0.111)
0
RXX 0.793
1
0
RXX 0.793
1
ZZ (0.534)
ZZ (0.534)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

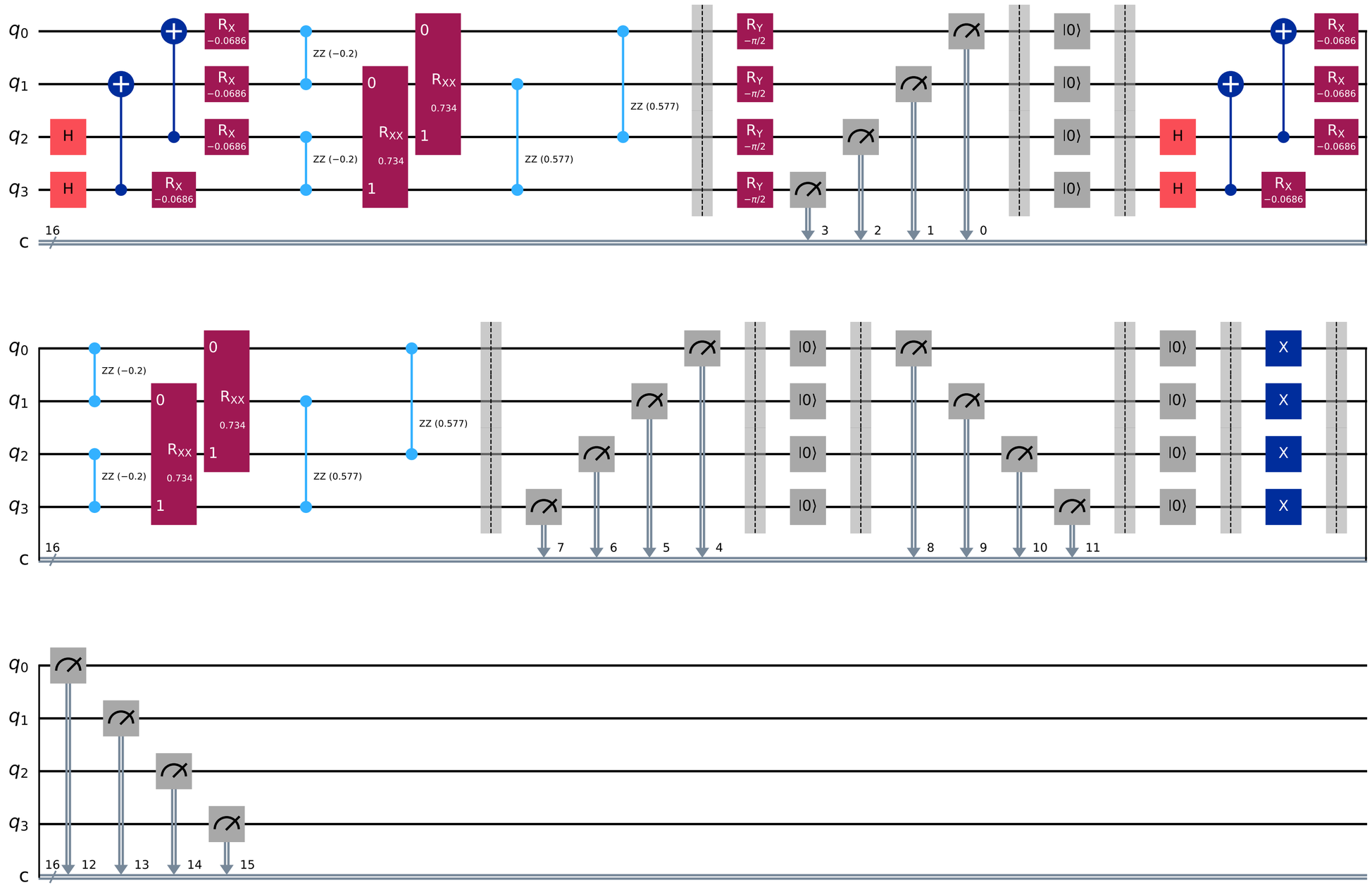

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.296
R_X −0.296
R_X −0.296
R_X −0.296
ZZ (−0.526)
ZZ (−0.526)
0
R_{XX} 0.766
1
ZZ (0.543)
ZZ (0.543)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.296
R_X −0.296
R_X −0.296
R_X −0.296
ZZ (−0.526)
ZZ (−0.526)
0
R_{XX} 0.766
1
ZZ (0.543)
ZZ (0.543)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

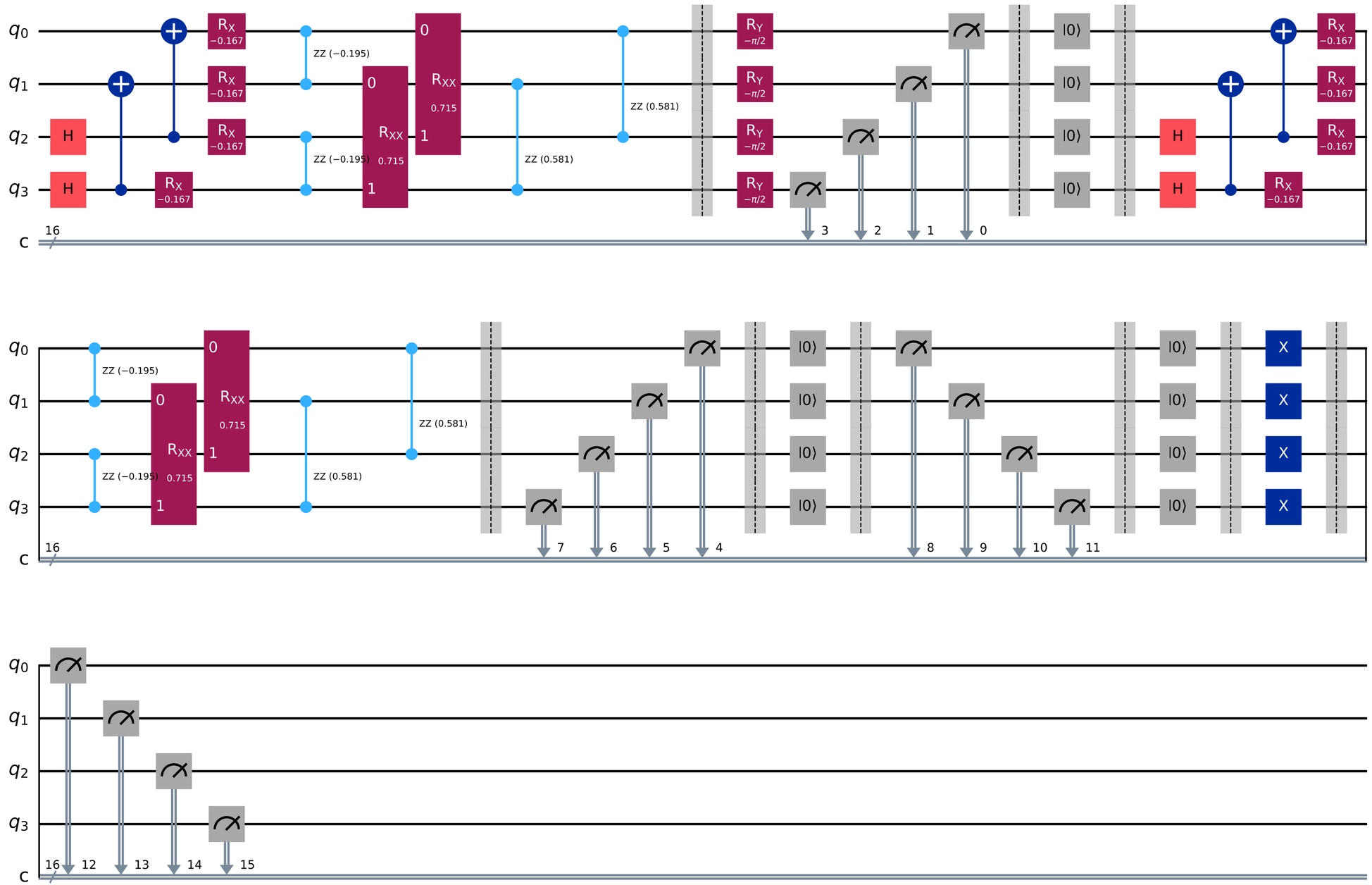

q0
q1
q2
q3
c
16
H
H
RX −0.193
RX −0.193
RX −0.193
RX −0.193
ZZ (0.0682)
ZZ (0.0682)
0
RXX
0.776
1
0
RXX
0.776
1
ZZ (0.54)
ZZ (0.54)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.193
RX −0.193
RX −0.193
RX −0.193
ZZ (0.0682)
ZZ (0.0682)
0
RXX
0.776
1
0
RXX
0.776
1
ZZ (0.54)
ZZ (0.54)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

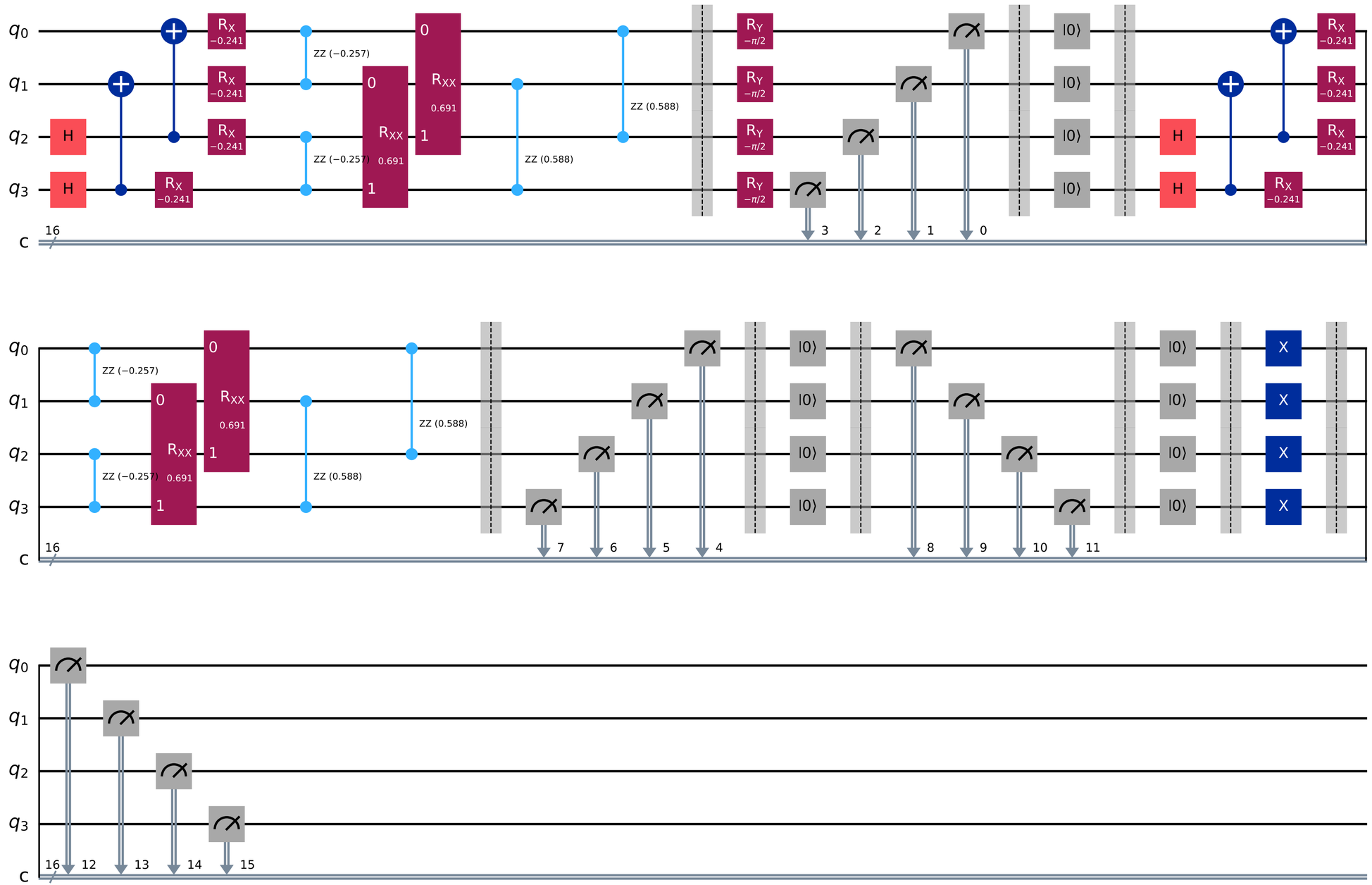

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0144
R_X −0.0144
R_X −0.0144
R_X −0.0144
ZZ (−0.111)
ZZ (−0.111)
0
R_{XX}
0.743
1
0
R_{XX}
0.743
1
ZZ (0.534)
ZZ (0.534)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0144
R_X −0.0144
R_X −0.0144
R_X −0.0144
ZZ (−0.111)
ZZ (−0.111)
0
R_{XX}
0.743
1
0
R_{XX}
0.743
1
ZZ (0.534)
ZZ (0.534)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

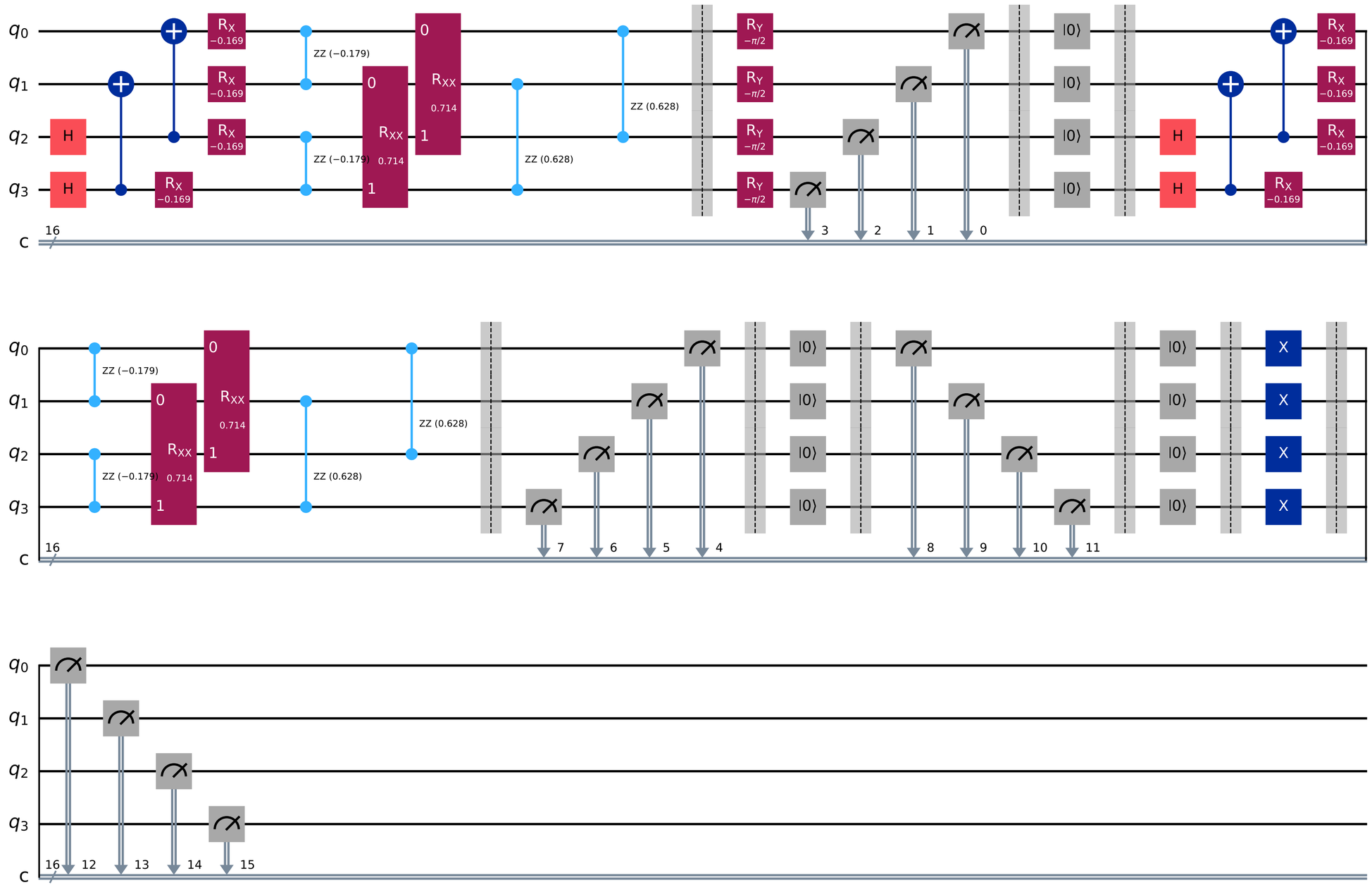

q0
q1
q2
q3
c
16
H
H
RX 0.085
RX 0.085
RX 0.085
RX 0.085
ZZ (−0.12)
ZZ (−0.12)
0
RXX 0.742
1
0
RXX 0.742
1
ZZ (0.539)
ZZ (0.539)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.085
RX 0.085
RX 0.085
RX 0.085
ZZ (−0.12)
ZZ (−0.12)
0
RXX 0.742
1
0
RXX 0.742
1
ZZ (0.539)
ZZ (0.539)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

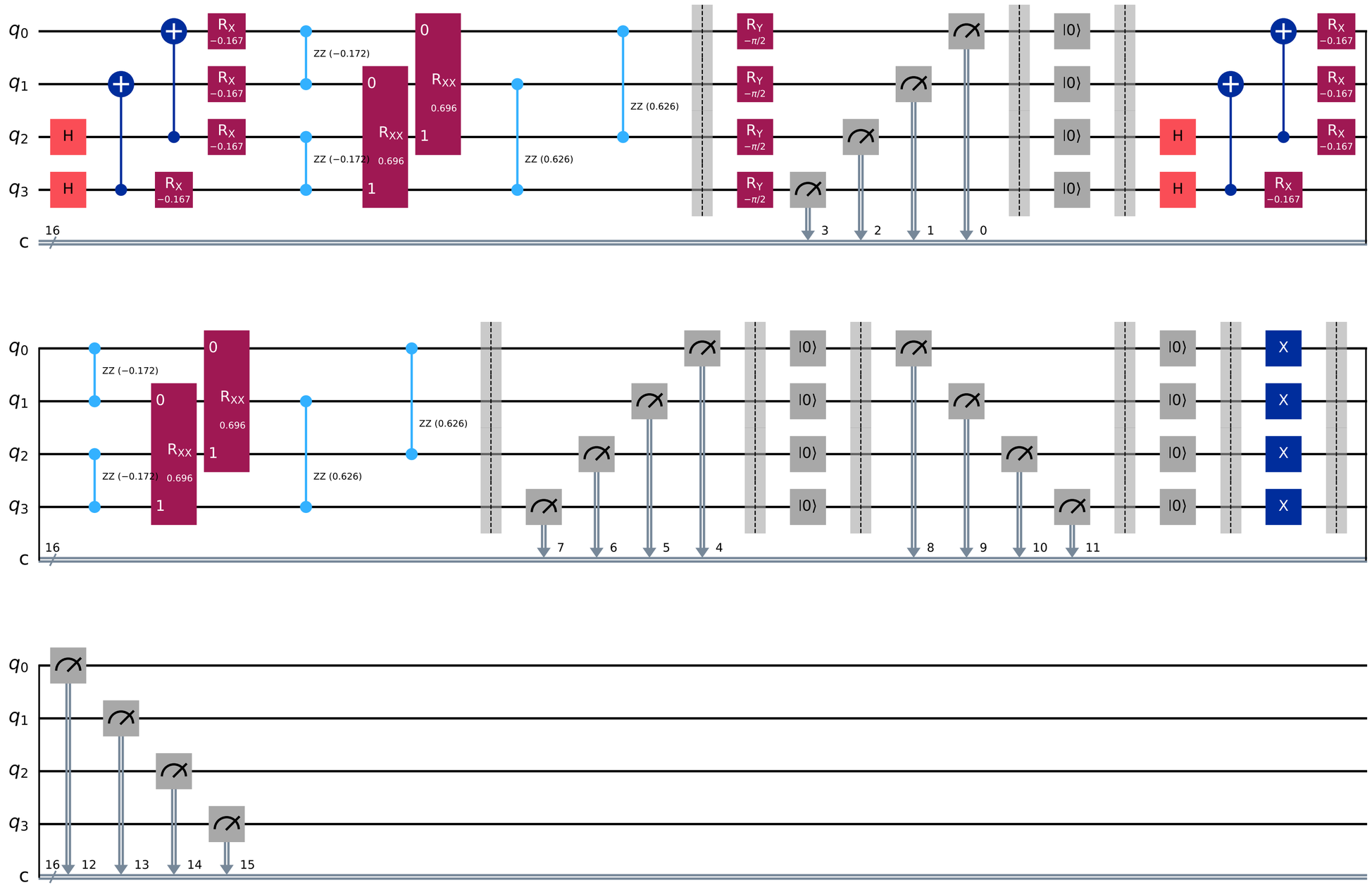

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0174
R_X −0.0174
R_X −0.0174
R_X −0.0174
ZZ (−0.115)
ZZ (−0.115)
0
R_{XX}
0.743
1
0
R_{XX}
0.743
1
ZZ (0.584)
ZZ (0.584)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0174
R_X −0.0174
R_X −0.0174
R_X −0.0174
ZZ (−0.115)
ZZ (−0.115)
0
R_{XX}
0.743
1
0
R_{XX}
0.743
1
ZZ (0.584)
ZZ (0.584)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

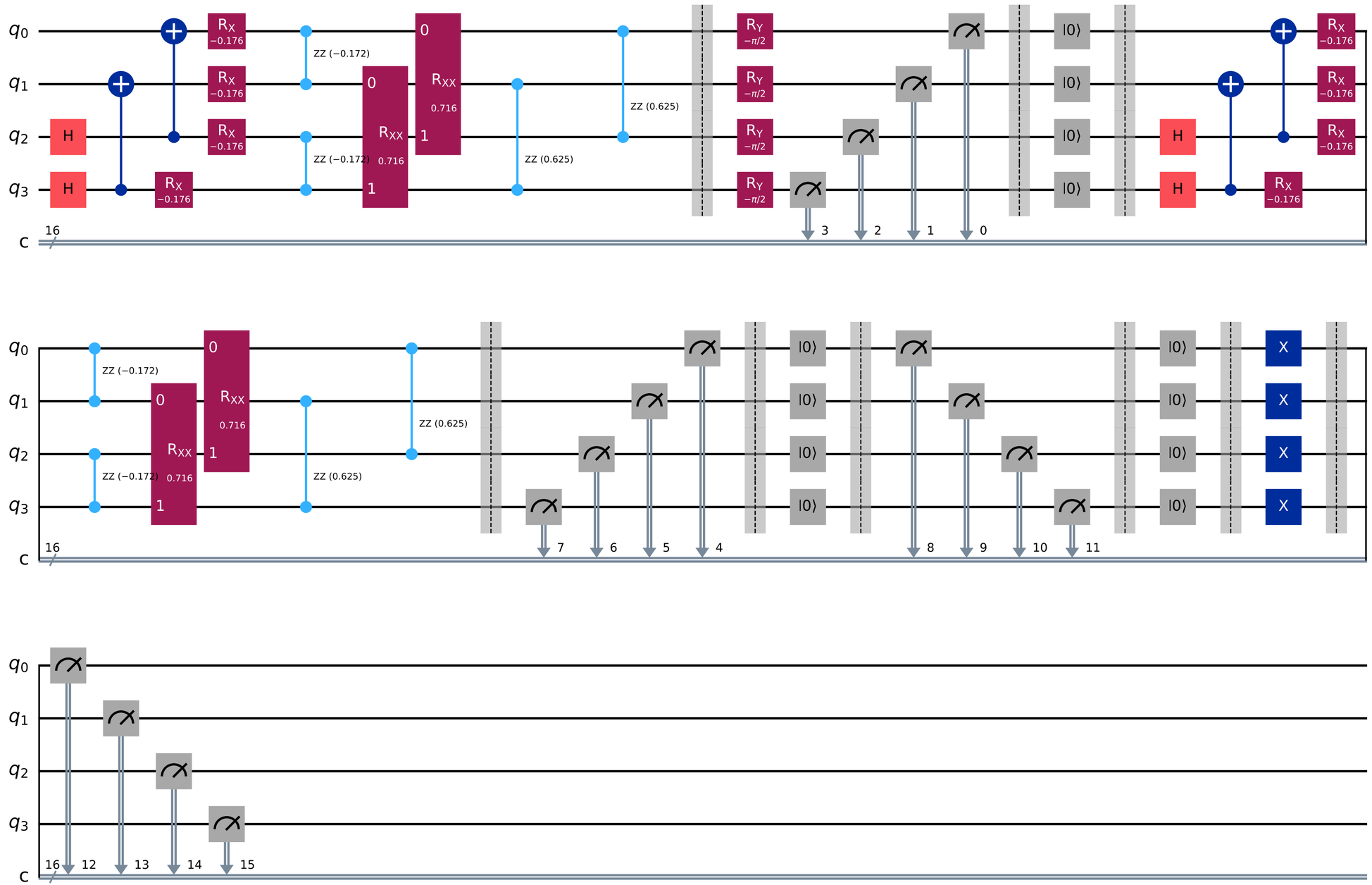

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0686
ZZ (−0.2)
R_{XX} 0.734
ZZ (0.577)
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

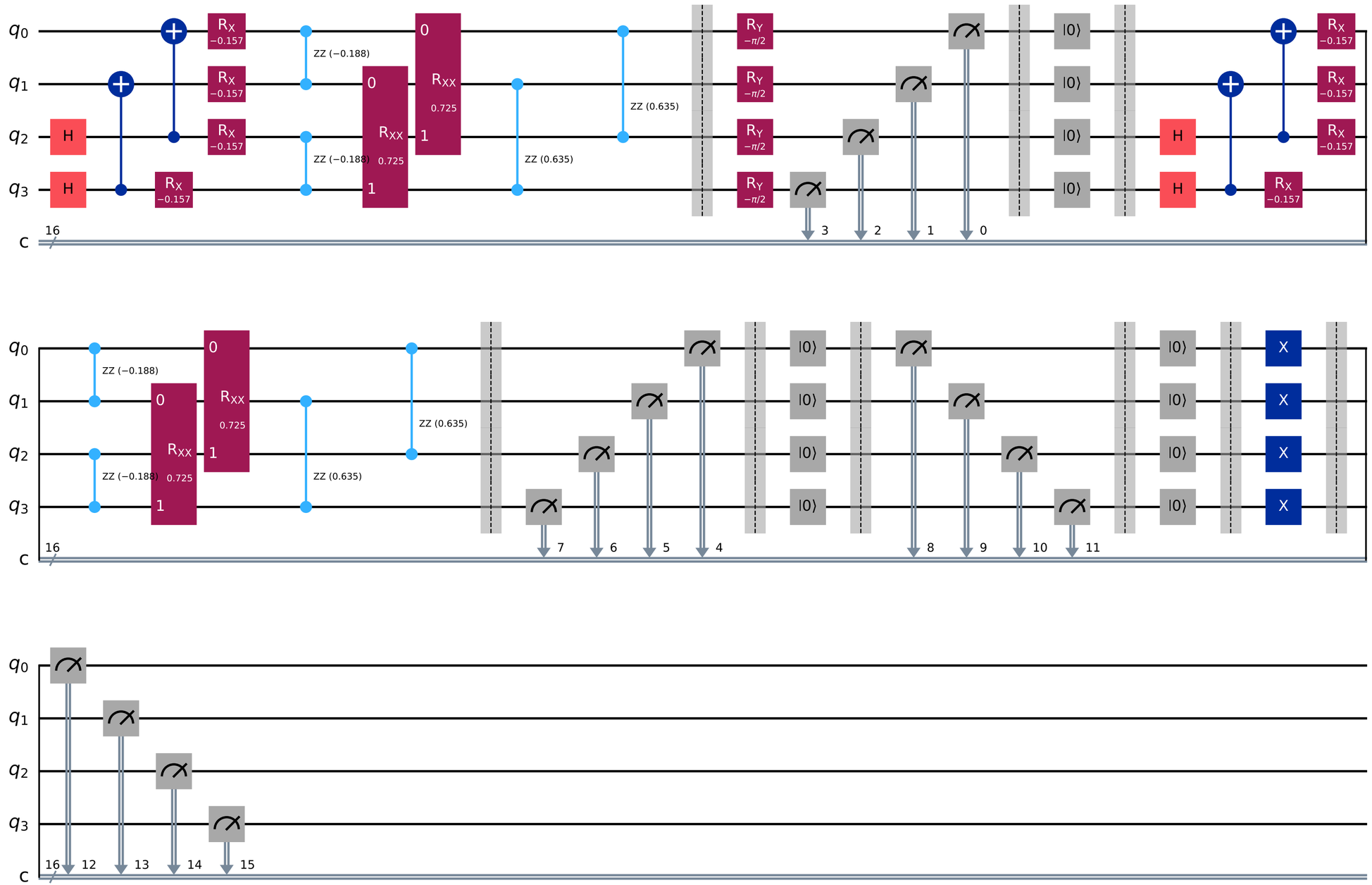

q0
q1
q2
q3
c
16
H
H
RX −0.167
RX −0.167
RX −0.167
RX −0.167
ZZ (−0.195)
ZZ (−0.195)
0
RXX 0.715
1
ZZ (0.581)
ZZ (0.581)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.167
RX −0.167
RX −0.167
RX −0.167
ZZ (−0.195)
ZZ (−0.195)
0
RXX 0.715
1
ZZ (0.581)
ZZ (0.581)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

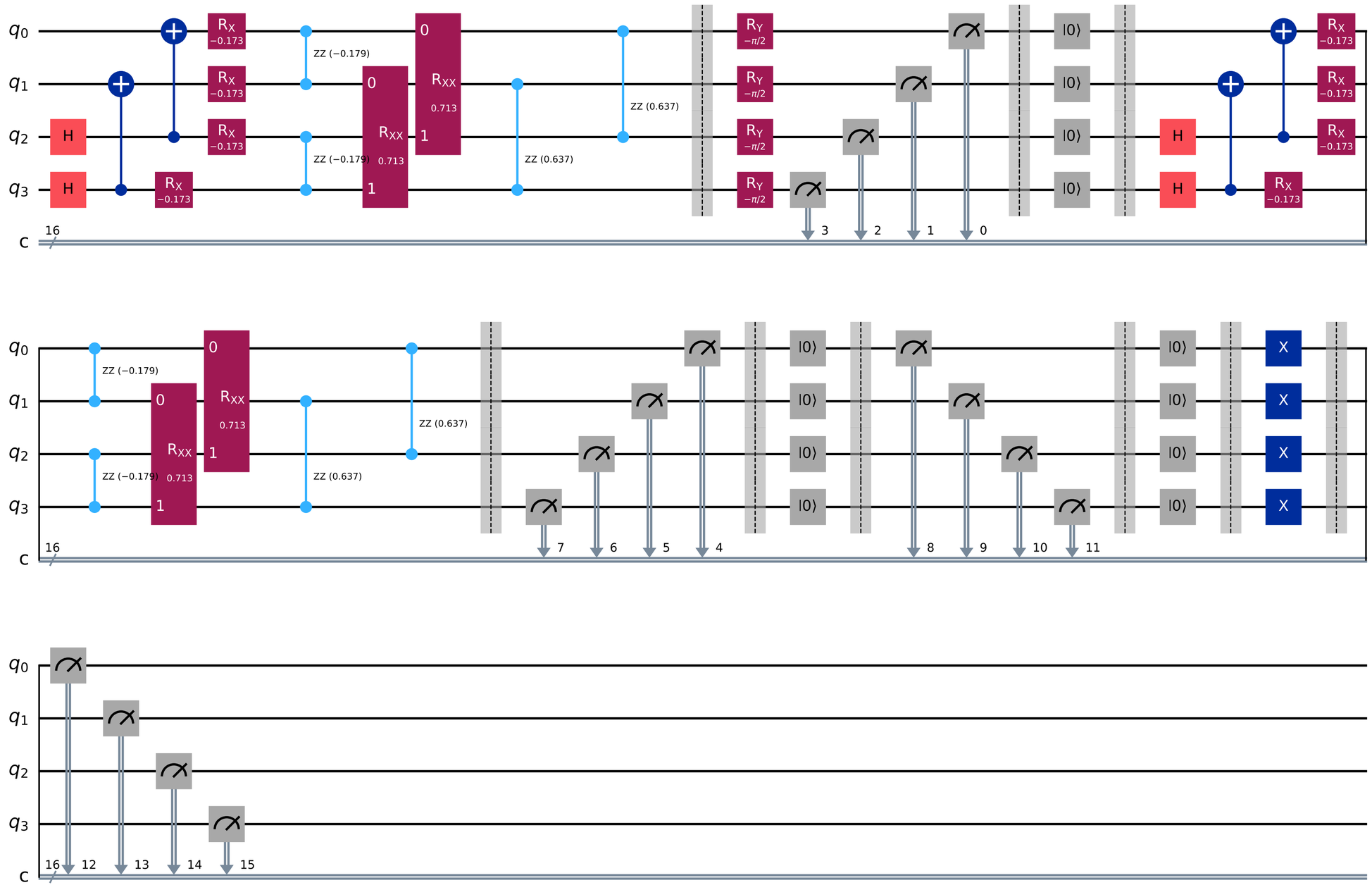

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.241
R_X −0.241
R_X −0.241
R_X −0.241
ZZ (−0.257)
ZZ (−0.257)
R_{XX} 0.691
ZZ (0.588)
ZZ (0.588)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.241
R_X −0.241
R_X −0.241
R_X −0.241
ZZ (−0.257)
ZZ (−0.257)
R_{XX} 0.691
ZZ (0.588)
ZZ (0.588)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

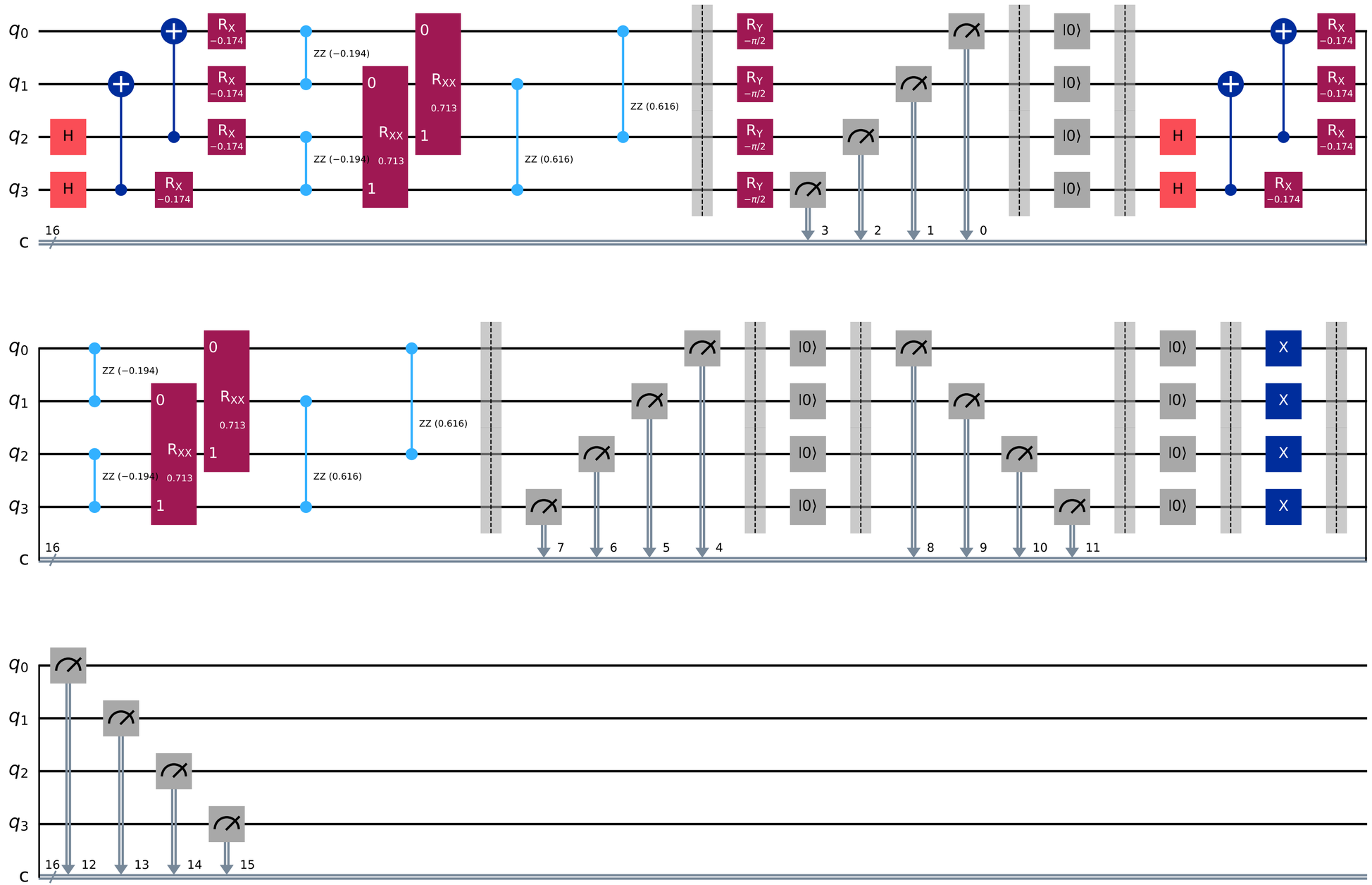

q0
q1
q2
q3
c
16
H
H
RX −0.169
RX −0.169
RX −0.169
RX −0.169
ZZ (−0.179)
ZZ (−0.179)
0
RXX 0.714
1
ZZ (0.628)
ZZ (0.628)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.169
RX −0.169
RX −0.169
ZZ (−0.179)
ZZ (−0.179)
0
RXX 0.714
1
ZZ (0.628)
ZZ (0.628)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.167
RX −0.167
RX −0.167
RX −0.167
ZZ (−0.172)
ZZ (−0.172)
0
RXX 0.696
1
0
RXX 0.696
1
ZZ (0.626)
ZZ (0.626)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.167
RX −0.167
RX −0.167
RX −0.167
ZZ (−0.172)
ZZ (−0.172)
0
RXX 0.696
1
0
RXX 0.696
1
ZZ (0.626)
ZZ (0.626)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.176
R_X −0.176
R_X −0.176
R_X −0.176
ZZ (−0.172)
ZZ (−0.172)
R_{XX} 0.716
ZZ (0.625)
ZZ (0.625)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.176
R_X −0.176
R_X −0.176
R_X −0.176
ZZ (−0.172)
ZZ (−0.172)
R_{XX} 0.716
ZZ (0.625)
ZZ (0.625)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.157
R_X −0.157
R_X −0.157
R_X −0.157
ZZ (−0.188)
ZZ (−0.188)
0
R_{XX} 0.725
1
ZZ (0.635)
ZZ (0.635)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.157
R_X −0.157
R_X −0.157
R_X −0.157
ZZ (−0.188)
ZZ (−0.188)
0
R_{XX} 0.725
1
ZZ (0.635)
ZZ (0.635)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.173
R_X −0.173
R_X −0.173
R_X −0.173
ZZ (−0.179)
ZZ (−0.179)
0
R_{XX} 0.713
1
ZZ (0.637)
ZZ (0.637)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.173
R_X −0.173
R_X −0.173
R_X −0.173
ZZ (−0.179)
ZZ (−0.179)
0
R_{XX} 0.713
1
ZZ (0.637)
ZZ (0.637)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.174
RX −0.174
RX −0.174
RX −0.174
ZZ (−0.194)
ZZ (−0.194)
0
RXX 0.713
1
ZZ (0.616)
ZZ (0.616)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.174
RX −0.174
RX −0.174
RX −0.174
ZZ (−0.194)
ZZ (−0.194)
0
RXX 0.713
1
ZZ (0.616)
ZZ (0.616)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15